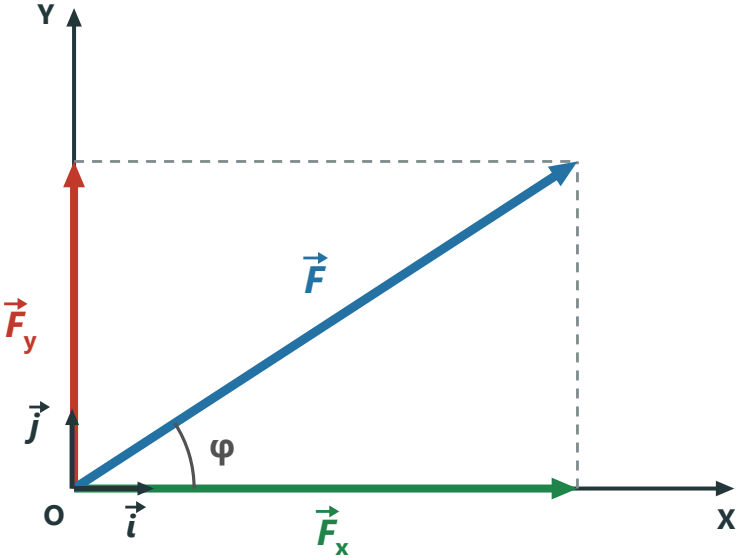

$$F = F_x \cdot i + F_y \cdot j$$

$$F_x = F \cdot \cos \varphi$$

$$F_y = F \cdot \operatorname{sen} \varphi$$

$$F = \sqrt{(F_x^{\,2} + F_y^{\,2})}$$

$$\operatorname{tg} \varphi = F_y / F_x$$